

UNIVERSITY OF VICTORIA, BC, Canada

WMO: 717830

Lat: 48.4569N

Lon: 123.3047W

Elev: 197

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.6	31.1	12.1	10.3	31.2	18.4	14.0	35.1	16.0	40.0	13.9	40.2	5.8	30	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	80.7	64.4	77.3	63.3	74.2	62.1	66.4	77.0	64.7	74.2	63.2	71.9	4.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	85.6	69.5	61.0	80.7	67.7	59.6	76.7	66.4	31.1	77.1	29.8	74.4	28.7	72.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.6	9.4	DB	22.9	87.5	4.2	3.6	19.9	90.0	17.4	92.1	15.1	94.1	12.0	96.7	(4)
(5)			WB	21.1	68.7	4.8	2.3	17.7	70.4	14.9	71.7	12.2	73.0	8.8	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	42.6	42.6	45.3	49.3	54.8	59.3	63.5	63.4	58.7	51.2	45.6	41.9		
	DBStd	8.99	5.08	4.66	4.28	4.08	4.49	4.09	3.90	3.54	3.96	3.87	4.95	5.15		
	HDD50	1093	233	207	154	60	9	0	0	0	0	32	143	255		
	HDD65	4971	696	626	611	472	318	179	75	74	193	429	581	717		
	CDD50	1662	2	1	8	37	157	279	420	415	262	67	12	2		
	CDD65	68	0	0	0	0	1	8	30	25	4	0	0	0		
	CDH74	726	0	0	0	0	17	103	320	248	38	0	0	0		
	CDH80	127	0	0	0	0	1	20	67	36	3	0	0	0		
(14) Wind	WSAvg	4.0	4.5	4.6	4.5	4.4	4.0	4.0	3.6	3.3	3.1	3.4	3.9	4.3		
(15) Precipitation	PrecAvg	25.4	4.2	2.9	1.9	1.3	1.0	0.7	0.6	0.8	1.3	2.4	3.9	4.4		
	PrecMax	34.4	8.5	6.7	4.3	2.8	2.0	2.7	1.5	2.6	2.8	8.1	7.5	10.4		
	PrecMin	15.3	0.6	0.4	0.5	0.7	0.2	0.1	0.0	0.0	0.2	0.2	0.6	0.4		
	PrecStd	4.9	2.1	1.5	1.1	0.6	0.5	0.6	0.4	0.7	0.7	1.8	1.7	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	54.3	60.6	68.6	76.4	83.1	86.5	84.1	78.4	66.4	58.0	55.8		
		MCWB	52.2	49.0	51.8	55.2	59.9	64.2	67.8	64.9	62.6	57.7	55.0	54.4		
	2%	DB	52.2	51.9	56.1	63.0	71.4	76.6	81.0	79.8	73.9	62.4	55.5	52.4		
		MCWB	50.3	47.9	49.9	53.5	58.3	62.3	65.2	64.4	61.4	55.6	53.3	50.4		
	5%	DB	50.4	50.2	53.7	59.4	67.4	72.1	77.3	76.5	70.5	59.6	53.6	50.2		
		MCWB	49.0	47.1	48.6	51.5	57.0	60.5	63.6	63.7	60.5	54.6	51.8	48.4		
	10%	DB	48.9	48.6	51.7	56.6	64.0	68.4	74.0	73.3	67.2	57.3	52.1	48.4		
		MCWB	47.6	46.1	47.7	50.0	55.3	58.5	62.3	62.4	59.3	53.5	50.5	46.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	51.3	53.8	57.6	62.1	66.1	69.7	68.2	65.3	61.9	56.5	55.0		
		MCDB	54.0	52.2	57.1	64.7	70.3	79.3	81.9	78.4	74.1	63.4	57.1	55.4		
	2%	WB	51.1	49.3	51.5	54.8	59.7	63.3	66.9	66.2	63.5	57.8	53.9	51.2		
		MCDB	51.7	50.6	54.2	61.1	68.1	74.2	78.4	75.9	70.7	59.9	54.7	51.9		
	5%	WB	49.4	48.0	49.9	53.0	57.9	61.3	64.9	64.6	61.8	55.7	52.2	48.9		
		MCDB	50.0	49.2	52.5	58.0	65.7	70.3	74.8	73.9	68.0	57.7	53.1	49.7		
	10%	WB	47.8	46.7	48.3	51.2	56.2	59.5	63.0	63.1	60.1	54.0	50.7	47.1		
		MCDB	48.6	48.1	51.0	55.3	62.8	67.0	72.1	71.6	65.4	56.2	51.7	48.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	9.0	11.1	13.5	16.2	17.1	19.9	19.2	16.0	11.3	8.6	7.5		
		MCDBR	8.1	9.9	13.1	17.9	22.6	24.2	26.4	25.3	21.6	14.3	9.1	9.0		
	5% WB	MCWBR	6.6	7.1	8.3	10.1	11.7	11.6	12.2	11.8	10.8	9.0	7.2	7.7		
		MCWBR	7.7	8.6	11.9	16.2	20.2	22.1	24.1	22.9	18.6	12.1	8.7	8.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.318	0.317	0.331	0.358	0.377	0.355	0.348	0.353	0.339	0.331	0.325	0.314		
		taud	2.526	2.550	2.525	2.441	2.402	2.482	2.539	2.522	2.561	2.565	2.507	2.490		
	Ebn at Noon	Ebn at Noon	241	270	283	284	280	287	288	282	277	260	234	225		
		Edh at Noon	19	23	28	33	36	33	31	30	26	23	19	18		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	295	575	899	1361	1741	1904	2079	1753	1249	687	353	237		
	RadStd	RadStd	39	77	105	127	137	180	145	113	102	76	54	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-1.10	-2.02	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page